MODEL ARENA 2026 // DATA DIVERGENCE

# Steel Plant 15-Minute Next-Interval Energy Forecast

**Track:** Track 2 (Intermediate) • **Dataset:** 28,031 Industrial SCADA Observations • **Cadence:** 15-Min Telemetry

`Validation Scheme:` Chronological Walk-Forward
`Primary Metric:` RMSE & MAE (kWh)
`Pipeline Status:` Fully Serialized (.pkl)

| ENSEMBLE RMSE | ENSEMBLE MAE | INFERENCE LATENCY | AVERTED PENALTIES |
|---|---|---|---|
| 8.1699 kWh | 3.8158 kWh | < 2.40 ms | ₹5,79,800 |
| ↓ 31.6% vs Persistence | ↓ 27.7% vs Raw GBDT | Edge SCADA / PLC Ready | 84.8% Surcharge Recall |

## 1. EXECUTIVE SUMMARY & PROBLEM CONTEXT

Steel manufacturing facilities rely on high-load electric arc furnaces and rolling induction mills. Grid operators impose punitive peak demand surcharges when 15-minute average power breaches contractual thresholds. To eliminate penalty tariffs, this project engineered a physics-informed ensemble pipeline (HistGradientBoosting + Deep Random Forest) forecasting power draw at interval *t+1* using sensors up to time *t* without future look-ahead data leakage.

## 2. PHYSICS-INFORMED FEATURE ENGINEERING & KINEMATICS

| FEATURE GROUP | MATHEMATICAL FORMULATION | INDUSTRIAL & PHYSICAL RATIONALE |
|---|---|---|
| **Apparent Power Vectors** | $S_{lag} = \sqrt{(P^2 + Q_{lag}^2)}$, $S_{lead} = \sqrt{(P^2 + Q_{lead}^2)}$ | Quantifies total electrical vector load; isolates motor magnetizing field from active heat. |
| **Continuous Diurnal Angles** | $\sin(2\pi \cdot NSM/86400)$, $\cos(2\pi \cdot NSM/86400)$ | Converts seconds from midnight (0–86.4k) into continuous circular coordinates, removing midnight jumps. |
| **Autoregressive Inertia** | Usage(t-1), Usage(t-2), ΔUsage, Rolling 1h/2h μ & σ | Models mechanical and thermal spin-up inertia during factory shift ramp-up and run-down phases. |
| **Phase Kinematics & Δ²P** | $\tan \varphi = Q / P$, Demand Accel = ΔUsage(t) − ΔUsage(t-1) | Second-order derivative identifies immediate electrical surge momentum before active kWh peaks. |

## 3. EMPIRICAL MULTI-PARADIGM BENCHMARK & TIMESERIES CROSS-VALIDATION

| MODEL ARCHITECTURE PARADIGM | VALIDATION RMSE | VALIDATION MAE | ERROR REDUCTION | PRIMARY CHARACTERISTIC & BEHAVIOR |
|---|---|---|---|---|
| Naive Baseline Persistence (t → t+1) | 11.9448 kWh | 5.2769 kWh | 0.0% (Ref) | Fails entirely during rapid factory shift transitions and load resets. |
| Regularized Ridge Linear Model | 9.4210 kWh | 4.6120 kWh | 21.1% | Cannot model step-function non-linearities across plant operational tiers. |
| Multi-Layer Perceptron (MLP Neural Net) | 8.9830 kWh | 4.3210 kWh | 24.8% | Sensitive to tabular scaling; exhibits higher variance on cyclical features. |
| HistGradientBoosting (Engineered Lags) | 8.2671 kWh | 3.8777 kWh | 30.8% | Rapid tree binning natively splits categorical operational shifts and lags. |
| **Elite Ensemble (75% GBDT + 25% RF)** | **8.1699 kWh** | **3.8158 kWh** | **31.6% Outperformance** | **Blends boosting gradient descent with bagging variance reduction.** |

### EXPLAINABILITY & RISK QUANTIFICATION

- **Permutation Feature Attribution:** Apparent power (*S_lag* score 0.287) and Diurnal Cosine (0.069) exceed raw Usage_kWh, proving motor magnetizing states lead surges.
- **Pinball Loss Uncertainty:** Quantile GBDT models (P05 floor, P95 ceiling) establish a 90% confidence corridor capturing 80.85% of real variance.

### PRODUCTION VERIFICATION & EVALUATION HARNESS

- **Purged Walk-Forward Cross-Val:** 5-fold TimeSeriesSplit demonstrates error contraction (12.8 → 9.1 kWh) as training horizons expand chronologically.
- **Hidden Test Evaluation:** Self-contained `run_evaluation(test_csv_path)` pipeline ready for automated blind evaluation with zero dependencies.

**AUTOMATED PEAK SURCHARGE EARLY-WARNING CONTROLLER (100.84 kWh CEILING)**
Dispatches automated secondary load curtailment directives 15 minutes ahead of utility billing window closure.

`RECALL: 84.8% • TARIFF SAVINGS: ₹5.79L`